Solucionario del Examen Sustitutorio CF1B2- FI-III - A y B 2023-1

P1:

$$\rho = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$$

Carga Total: $Q = \int \rho(r)\, dV$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{R} \rho_0 \left(1 - \frac{r^2}{R^2}\right) r^2 \operatorname{sen}\theta\, d\theta\, d\phi\, dr$$

$$= 4\pi\rho_0 \int_0^{R} \left(r^2 - \frac{r^4}{R^2}\right) dr$$

$$= 4\pi\rho_0 \left\{ \frac{r^3}{3}\Big|_0^R - \frac{r^5}{R^2}\Big|_0^R \right\}$$

$$= 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^3}{5}\right] = 4\pi\rho_0 R^3 \left[\frac{1}{3} - \frac{1}{5}\right]$$

$$Q = \frac{8\pi}{15}\rho_0 R^3$$

El campo eléctrico se obtiene aplicando Gauss en todos los puntos del espacio.

Caso $r \geq R$, la superficie gaussiana será exterior a la esfera, esto es un cascarón de radio $r > R$. para el cual se cumple $\iint_S \bar{E}\cdot d\bar{S} = \frac{q_{encerrada}}{\epsilon_0}$

$$4\pi r^2 E(r) = \frac{1}{\epsilon_0}\,\frac{8\pi}{15}\rho_0 R^3$$

$$E(r) = \frac{2}{15}\frac{\rho_0}{\epsilon_0}\frac{R^3}{r^2}\hat{r}$$

Caso $r < R$, aquí solo se debe considerar la carga dentro del cascarón.

$$4\pi r^2 E(r) = \frac{q_{enc}}{\epsilon_0} = \frac{1}{\epsilon_0}\int_0^{2\pi}\int_0^{\pi}\int_0^{r} \rho_0\left(1 - \frac{r^2}{R^2}\right) r^2 \operatorname{sen}\theta\, dr\, d\theta\, d\phi$$

$$4\pi r^2 E(r) = \frac{4\pi\rho_0}{\epsilon_0}\left(\frac{r^3}{3} - \frac{r^5}{5R^2}\right)$$

$$E(r) = \frac{\rho_0}{\epsilon_0}\left(\frac{r}{3} - \frac{r^3}{5R^2}\right)\hat{r}$$

La energía potencial en todo el espacio: $W = \frac{\epsilon_0}{2} \int_{\text{todo el espacio}} |E^2| dV$

$$W(r>R) = \frac{\epsilon_0}{2} \int_0^{2\pi}\int_0^{\pi}\int_R^{\infty} \left[\frac{2}{15}\frac{\rho_0}{\epsilon_0}\frac{R^3}{r^2}\right]^2 r^2 \operatorname{sen}\theta \, d\theta \, dr \, d\varphi$$

$$= \frac{\epsilon_0}{2}(4\pi)\left[\frac{2}{15}\frac{\rho_0 R^3}{\epsilon_0}\right]^2 \int_R^{\infty} \frac{1}{r^4} r^2 dr$$

$$W(r>R) = \frac{8\pi \rho_0^2 R^6}{15^2 \epsilon_0} \int_R^{\infty} \frac{dr}{r^2} = \frac{8\pi \rho_0^2 R^6}{15^2 \epsilon_0}\left(-1\right)\frac{1}{r}\Big|_R^{\infty}$$

$$= \frac{8\pi \rho_0^2 R^6}{15^2 \epsilon_0} \times \frac{1}{R} = \frac{8\pi \rho_0^2}{3\times 15^2 \epsilon_0} R^5$$

$$W(r<R) = \frac{\epsilon_0}{2} \int_0^{2\pi}\int_0^{\pi}\int_0^{R} \left[\frac{\rho_0}{\epsilon_0}\left(\frac{r}{3} - \frac{r^3}{5R^2}\right)\right]^2 r^2 dr \operatorname{sen}\theta \, d\theta \, d\varphi$$

$$= \frac{\epsilon_0}{2}(4\pi)\frac{\rho_0^2}{\epsilon_0^2}\int_0^R \left(\frac{r}{3} - \frac{r^3}{5R^2}\right)^2 r^2 dr$$

$$= \frac{2\pi \rho_0^2}{\epsilon_0}\int_0^R \left[\frac{r^2}{9} + \frac{r^6}{25R^4} - 2\left(\frac{r}{3}\right)\left(\frac{r^3}{5R^2}\right)\right] r^2 dr$$

$$= \frac{2\pi \rho_0^2}{\epsilon_0}\left\{\frac{r^5}{45} + \frac{r^9}{25\times 9 R^4} - \frac{2}{15\times 7}\frac{r^7}{R^2}\right\}_0^R$$

$$= \frac{2\pi}{15}\frac{\rho_0^2}{\epsilon_0}\left\{\frac{R^5}{3} + \frac{R^5}{15} - \frac{2}{7}R^5\right\}$$

$$= \frac{2\pi \rho_0^2}{15 \epsilon_0} R^5 \times \frac{12}{7\times 15} = \frac{24\pi \rho_0^2 R^5}{7\times 15^2 \epsilon_0}$$

$$W_{\text{todo el espacio}} = \frac{8\pi \rho_0^2}{3\times 15^2 \epsilon_0} R^5 + \frac{24\pi \rho_0^2 R^5}{7\times 15^2 \epsilon_0} = \left(\frac{26}{51}\right)\frac{8\pi \rho_0^2 R^5}{15^2 \epsilon_0}$$

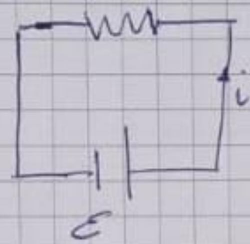
P#2:

(a) Si la superficie está orientada que entra hacia la figura, el flujo ϕ_B.

$$\phi_B = \iint_S \bar{B}\cdot d\bar{S} = \iint_S B\,dS = B\iint dS = BDz$$

$$\mathcal{E} = -\frac{d\phi_B}{dt} = -BD\frac{dz}{dt} = -BDv$$

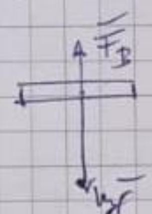
El circuito de corriente



$$\Rightarrow i = \frac{|\mathcal{E}|}{R} = \frac{BDv}{R}$$

(b) Si se parte del reposo $v(0)=0$

y la ecuación de movimiento para la barra



$$mg - F_B = ma = m\dot{v} \quad \text{--- (I)}$$

$$F_B = BDi = BD\left(\frac{BDv}{R}\right)$$

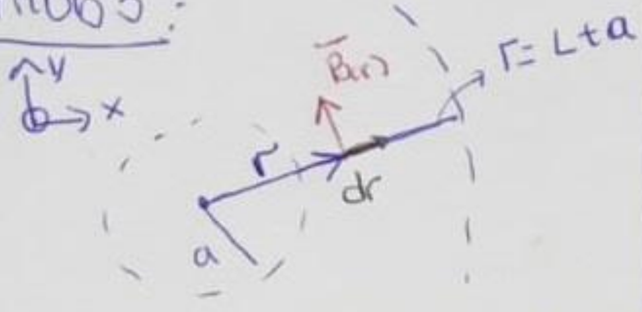
$$F_B = \frac{(BD)^2 v}{R}$$

De (I): $\quad \dot{v} + \dfrac{(BD)^2 v}{mR} = g$

Con la condición inicial $v(0)=0$

$$\boxed{v(t) = \frac{mgR}{(BD)^2}\left[1 - e^{-\frac{(BD)^2}{mR}t}\right]}$$

Prob3:



$$d\bar{F} = \overset{\circ}{I}\, d\bar{l} \times \bar{B} = \overset{\circ}{I} B\, dr\, \hat{k} \quad (1P)$$

$$1)\quad \bar{F} = I \int_a^{L+a} \frac{\alpha}{r^2}\, dr\ \hat{k} \quad (1P)$$

$$\boxed{\bar{F} = \frac{I\alpha L}{(L+a)a}\,\hat{k}} \quad (0{,}5P)$$

b) $d\bar{\tau} = \bar{r} \times d\bar{F}$

$$d\tau = r\,dF = \frac{I\alpha}{r^2}\, r\, dr \quad ..(1)$$

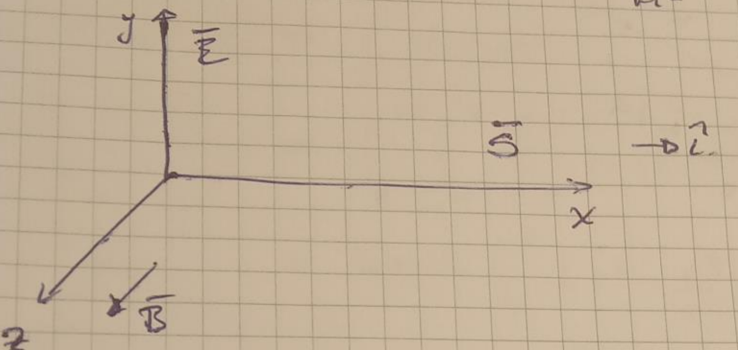
$$\int d\tau = I\alpha \int_a^{L+a} \frac{dr}{r} \quad .. \quad (0{,}5)$$

$$\bar{\tau} = I\alpha \ln\left(\frac{L+a}{a}\right)\left(\cos\theta\,\hat{i} - \sin\theta\,\hat{j}\right) \quad (1P)$$

P4:

$$\bar{S} = \frac{1}{\mu_0}\bar{E} \times \bar{B} = 100\ \frac{W}{m^2} \cos^2(kx - \omega t)\ \hat{i}$$

(a)



Se dirige hacia la dirección positiva del eje "x"

(b)

$k = 10\ \frac{rad}{m}$, $\quad \omega = 9 \times 10^9\ \frac{rad}{s}$

$\lambda = \frac{2\pi}{k}$ $\qquad f = \frac{\omega}{2\pi}$

$\lambda = 0{,}628\,m$ $\qquad f \cong 4{,}8 \times 10^{10}\ Hz$

(c)

$$\bar{E} = 10\sqrt{\mu_0}\ \cos(10x - 3\times 10^9 t)\ \hat{j}$$

$$\bar{B} = \frac{10\sqrt{\mu_0}}{c}\ \cos(10x - 3\times 10^9 t)\ \hat{k}$$